Project Report: Bike Price Prediction System

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1. Introduction

The goal of this project was to build a **Bike Price Predictor** using a machine learning approach. The system estimates the resale value of a bike based on factors such as its age, kilometers driven, engine power, and brand.

This project combines **data preprocessing and linear regression modeling** which i learnt in PHY1003 course to create a simple yet functional prediction system.

2. Problem Statement

Buying and selling used bikes often involves uncertainty in pricing. Sellers may overprice, while buyers may undervalue a vehicle due to lack of proper data.

Additionally, high-end bikes are often expensive, making them unaffordable for many people, especially students. Because of this, a large number of people prefer buying second-hand bikes as a more economical option.

This project is also inspired by a personal problem I faced while trying to buy a bike, where I struggled to determine whether the asking price was fair.

3. Why This Problem Matters

- Many students and budget buyers rely on second-hand bikes
- High prices of new bikes make resale markets more important
- Helps buyers avoid overpaying
- Assists sellers in setting realistic prices

- Introduces a data-driven approach instead of guesswork

4. Approach and Methodology

4.1 Data Collection

The dataset (bikes.csv) contains:

- Brand
- Age of the bike
- Kilometers driven
- Engine power (cc)
- Price

The dataset was sourced from Kaggle, which helped in understanding how to find and use real world datasets.

4.2 Data Preprocessing

Steps taken:

- Removed missing values using dropna()
- Converted categorical brand names into numeric IDs
- Normalized features using Min-Max Scaling

I also learned how CSV files work and how they can be used as datasets in Python using pandas.

4.3 Feature Engineering

Selected features:

- Age
- Kilometers driven
- Engine power

- Brand

4.4 Model Selection

A Linear Regression model was implemented using the Normal Equation.

Initially, I faced difficulty in:

- Understanding regression concepts
- Applying them in a real Python program

However, during this course, I was able to:

- Understand the mathematical concept of regression
- Successfully implement it in code

4.6 Evaluation Metrics

Two metrics were used:

- Mean Absolute Error (MAE)
  Measures average prediction error in rupees
- R^2 Score (Coefficient of Determination)
  Indicates how well the model explains variance in price

5. Key Design Decisions

1. Manual Implementation of Regression

- Helped in deeply understanding how regression works
- Avoided dependency on external libraries

2. Fixed Mileage Growth

- Assumed ~8000 km/year
- Simplifies predictions but may not reflect real-world variation

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7. Challenges Faced

1. Understanding Regression in Practice

- Initially difficult to connect theory with implementation
- Faced issues while applying formulas in Python

2. Working with CSV Data

- Learned how to read and manipulate CSV files using pandas
- Understood how datasets are structured and used in ML

4. Dataset Limitations

- Limited features (no condition, location, etc.)

8. Results

- The model predicts approximate bike prices
- MAE provides an estimate of prediction error
- R^2 score shows moderate performance

The system works effectively for a basic implementation.

9. What I Learned

Technical Learnings

- Implementation of linear regression from scratch
- Feature scaling using normalization
- Working with CSV datasets in Python
- Using Kaggle to find real-world datasets

Practical Learnings

- Real-world problems are often messy and require assumptions

- Simpler models can still provide useful insights
- Data plays a crucial role in prediction accuracy

10. Future Improvements

- Use advanced models
- Add more features (condition, ownership history, location)
- Add visualization features

11. Conclusion

This project demonstrates how machine learning can be applied to solve a real world problem faced by many people, especially students and budget buyers.

It also reflects a personal learning journey from struggling with regression concepts to successfully implementing a working prediction system.

Overall, the project provided valuable experience in:

- Regression Model building
- Problem-solving using machine learning

End of Report